

MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction

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Research Questions

- How can multimodal LLMs move beyond turn-based interaction to enable real-time full-duplex omni-modal communication?
- Can a unified streaming framework align multimodal inputs and outputs along a shared temporal axis for simultaneous perception and response?
- What architectural design enables proactive behavior while maintaining efficiency on edge devices with limited resources?

Core Idea

Current multimodal models suffer from alternating perception-response phases that prevent real-time adjustment and proactive behavior. MiniCPM-o 4.5 introduces Omni-Flow, a streaming framework that partitions continuous interaction into fine-grained time windows where the model simultaneously processes visual/audio inputs and generates text/speech outputs. The key insight is treating interaction as time-aligned streams rather than conversational turns, enabling the model to see, listen, and speak simultaneously while exhibiting proactive behaviors like scene commentary or reminders.

Key Formula

Length Reward Function

$r_{\text{len}}(i) = \begin{cases} s_i, & r_i = 1 \\ \min(0, s_i), & r_i = 0 \end{cases}$
Length reward for reinforcement learning where r_i is correctness indicator, s_i is scaled length penalty. The $\min(0, s_i)$ term prevents rewarding short incorrect responses while encouraging conciseness for correct ones.

Smooth Length Penalty

$$s_i = \left(0.5 - \frac{\ell_i - \ell_{\min}}{\ell_{\max} - \ell_{\min}}\right) \times \min\left(1, \frac{\ell_{\max} - \ell_{\min}}{\tau}\right)$$

Smooth length scaling where $\ell_i, \ell_{\min}, \ell_{\max}$ are response lengths and τ downscales reward when length differences are small. This avoids aggressive penalization that conflicts with accuracy rewards.

Key Figure

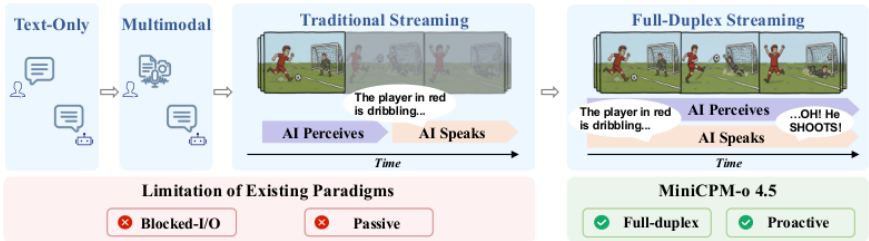


Figure 3: **From turn-based interaction to full-duplex streaming.** Existing interaction paradigms separate perception and response as alternating phases, leading to blocked information flow and passive behavior. In contrast, MiniCPM-o 4.5 continuously perceives incoming multimodal streams while speaking, allowing the model to update its response in real time and act proactively.

Figure 3: Illustrates the core paradigm shift from blocked turn-based interaction to full-duplex streaming where the model continuously perceives while speaking, enabling real-time response updates.

Limitations

- Speech generation in omni-modal streaming mode can be unstable with mispronunciation or unintended English-Chinese code-mixing
- Foundation capability and robustness in long, dynamic real-world streaming interactions require further improvement
- Proactive behavior remains relatively simple, lacking richer context-aware planning and self-initiated assistance
- Web demo may experience latency or missing output fragments under unstable network conditions